Solution:

(i) Let us use the pair $(1,2)$ to make the relation "not symmetric" and consider the relation $\{(1,2)\}$. It is transitive. If we include $(2,3)$ and not include $(1,3)$, then the relation is not transitive. So the relation $\{(1,2),(2,3)\}$ is not reflexive, not symmetric and not transitive. Similarly we can construct more examples.

(ii) Just now we have seen that the relation $\{(1,2)\}$ is transitive, not reflexive and not symmetric.

(iii) Let us start with the pair $(1,2)$. Since we need symmetricity, we have to include the pair $(2,1)$. At this stage as $(1,1),(2,2)$ are not here, the relation is not transitive. Thus $\{(1,2),(2,1)\}$ is not reflexive; it is symmetric; and it is not transitive.

(iv) If we include the pairs $(1,1)$ and $(2,2)$ to the relation discussed in (iii), it will become transitive. Thus $\{(1,2),(2,1),(1,1),(2,2)\}$ is not reflexive; it is symmetric and it is transitive.

(v) For a relation on $\{0,1,2,3\}$ to be reflexive, it must have the pairs $(0,0),(1,1),(2,2),(3,3)$. Fortunately, it becomes symmetric and transitive. Therefore, as in (i) if we insert $(1,2)$ and $(2,3)$ we get the required one. Thus $\{(0,0),(1,1),(2,2),(3,3),(1,2),(2,3)\}$ is reflexive; it is not symmetric and it is not transitive.

(vi) Proceeding like this we get the relation $\{(0,0),(1,1),(2,2),(3,3),(1,2)\}$ that is reflexive, transitive and not symmetric.

(vii) As above we get the relation $\{(0,0),(1,1),(2,2),(3,3),(1,2),(2,3),(2,1),(3,2)\}$ that is reflexive, symmetric and not transitive.

(viii) We have the relation $\{(0,0),(1,1),(2,2),(3,3)\}$ which is reflexive, symmetric and transitive.

Example 1.13 In the set $\mathbb{Z}$ of integers, define mRn if $m-n$ is a multiple of 12. Prove that R is an equivalence relation.

Solution:

As $m-m=0$ and $0=0\times 12$, hence mRm proving that R is reflexive.

Let mRn. Then $m-n=12k$ for some integer k; thus $n-m=12(-k)$ and hence nRm. This shows that R is symmetric.

Let mRn and nRp; then $m-n=12k$ and $n-p=12\ell$ for some integers k and ℓ.
So $m-p=12(k+\ell)$ and hence mRp. This shows that R is transitive.

Thus R is an equivalence relation.

Theorem 1.1: The number of relations from a set containing m elements to a set containing n elements is 2^{mn}. In particular the number of relations on a set containing n elements is 2^{n^2}.

Proof. Let A and B be sets containing m and n elements respectively. Then $A\times B$ contains mn elements and $A\times B$ has 2^{mn} subsets. Since every subset of $A\times B$ is a relation from A to B, there are 2^{mn} relations from a set containing m elements to a set containing n elements.

Taking $A=B$, we see that the number of relations on a set containing n elements is 2^{n^2}. ∎

(i) The number of reflexive relations on a set containing n elements is 2^{n^2-n}.

(ii) The number of symmetric relations on a set containing n elements is $2^{\frac{(n^2+n)}{2}}$.

Definition 1.5

If R is a relation from A to B, then the relation R^{-1} defined from B to A by

$$R^{-1}=\{(b,a):(a,b)\in R\}$$

is called the ***inverse*** of the relation R.